

PITSIGHT

BMW Customer Sentiment Intelligence

Turning 102,848 customer voices into actionable intelligence

90.7%

ML Prediction Accuracy

Predicting customer satisfaction
from operational data alone

102,848

Reviews Analyzed

25

BMW Models

6

Sentiment Aspects

7 Years

2019–2025

21

Countries

The Problem

Sales dashboards show **WHAT** happened.

They don't show WHY.

A model's sales are declining, but is it pricing, reliability, dealership experience, or technology?

Without analyzing customer voice at scale, **product teams are guessing.**

PitSight connects sales trends to customer
Sentiment:- Answering WHY, not just WHAT.

WHAT ?

Sales dropped 80% !!



WHY ?

Service visits rose 3x !!

Reason:

Maintenance score: 3.7 → 1.4

Technology score: 3.2 → 2.7

Satisfaction: 96% → 0%

How It Works

Three-stage intelligence pipeline

01 Data Engineering

AI-assisted web scraping for real market data

22,145 sales transactions across 21 countries

102,848 customer reviews generated

In-depth research on accuracy on models being sold regionally



02 NLP Scoring

6 aspects scored per review (1–5 scale)

Maintenance · Comfort · Technology

Service · Fuel/Range · Performance

Zero keyword overlap, EV/ICE separated



03 ML Prediction

Random Forest: 90.7% accuracy

Service History (45%) + Service Time (42%)

= 87% of prediction power

Demographics explain <10%

Core Finding: "How often it broke + How long the service took" = 87% of whether a customer is satisfied. Demographics, age, warranty, barely matter.

How NLP Scores a Review



From raw text → aspect scores → satisfaction label

STEP 1 Raw Customer Review

"Ride quality is **smooth and refined**, service center staff were **professional and efficient**, fuel efficiency is **impressive** for such car, workshop visits have been **minimal and predictable**, the tech in this car is **years ahead of competition**."

STEP 2 NLP extracts keywords → maps to 6 aspects → assigns scores (1–5)

Aspect	Keywords Found	Sentiment	Score
Maintenance	"minimal and predictable"	Positive	4
Comfort	"smooth and refined"	Positive	3
Technology	"years ahead of competition"	Positive	5
Service	"professional and efficient"	Positive	4
Fuel/Range	"impressive"	Positive	0
Performance	(not mentioned)	N/A	0



STEP 3 Calculate Sentiment Score

Average of non-zero scores only:

$(4 + 3 + 5 + 4) \div 4 = \mathbf{4.0}$ (scores of 0 = not mentioned, excluded)

PitSight

STEP 4 Apply Threshold (3.25)

$4.0 \geq 3.25 \rightarrow \mathbf{\text{Satisfied}}$

If $< 3.25 \rightarrow \mathbf{\text{Dissatisfied}}$

BMW
GROUP



Click here to follow

Portfolio Performance &

Customer Intelligence





>

Interactive landing page — each card links to its respective Power BI report



Portfolio Performance

Sales · Revenue · Market Trends
Country & Channel Analysis
Model-Level Performance

 [Click to Open](#)



Customer Intelligence

NLP Sentiment Analysis
Aspect-Level Scoring
Service & Satisfaction Diagnostics

 [Click to Open](#)

Sales

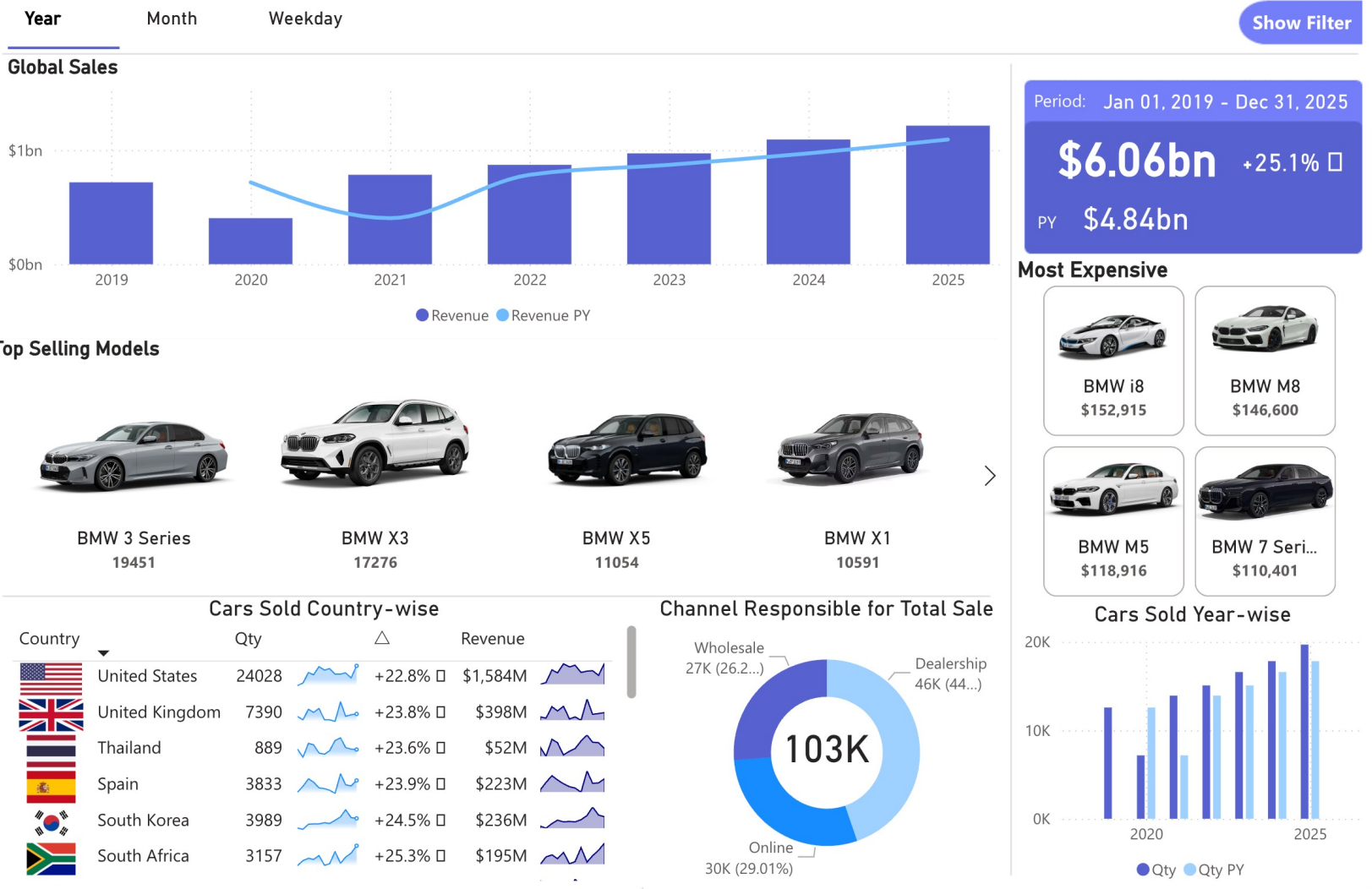
\$6.06bn revenue · 103K units · 25 models · 21 countries



Insights

Dashboard

All Models



Market Impact

Revenue grew 25% YoY (\$4.84bn→\$6.06bn)

Online channel surged from 2% to 29%

US dominates with 24K units but growth is slowing, Asia is the next frontier

Why It Matters

This page establishes the baseline:-
What sold, Where, and When.

But it can't explain WHY the sales are declining or WHY sales are constant/surging.

That's where NLP comes in picture

Models

Yearly performance, pricing, and sales trajectory for any selected model



Insights

Dashboard

All Models



BMW X6

Year	Qty	Revenue
2019	113	\$9M
2020	69	\$6M
2021	65	\$5M
2022	51	\$4M
2023	23	\$2M
2024	16	\$1M
2025	22	\$2M
Total	359	\$29M



\$82.0K

Average Price



BMW 1 Series
7454



BMW 2 Series
3578



BMW 3 Series
19451



BMW 4 Series
5928



BMW 5 Series
8603



BMW 7 Series
2049

Let’s begin with the example of BMW X6 -- The Warning Signs

Sales declined 80%

113 units (2019) → 22 units (2025)

Revenue collapsed

\$9M → \$2M in 6 years

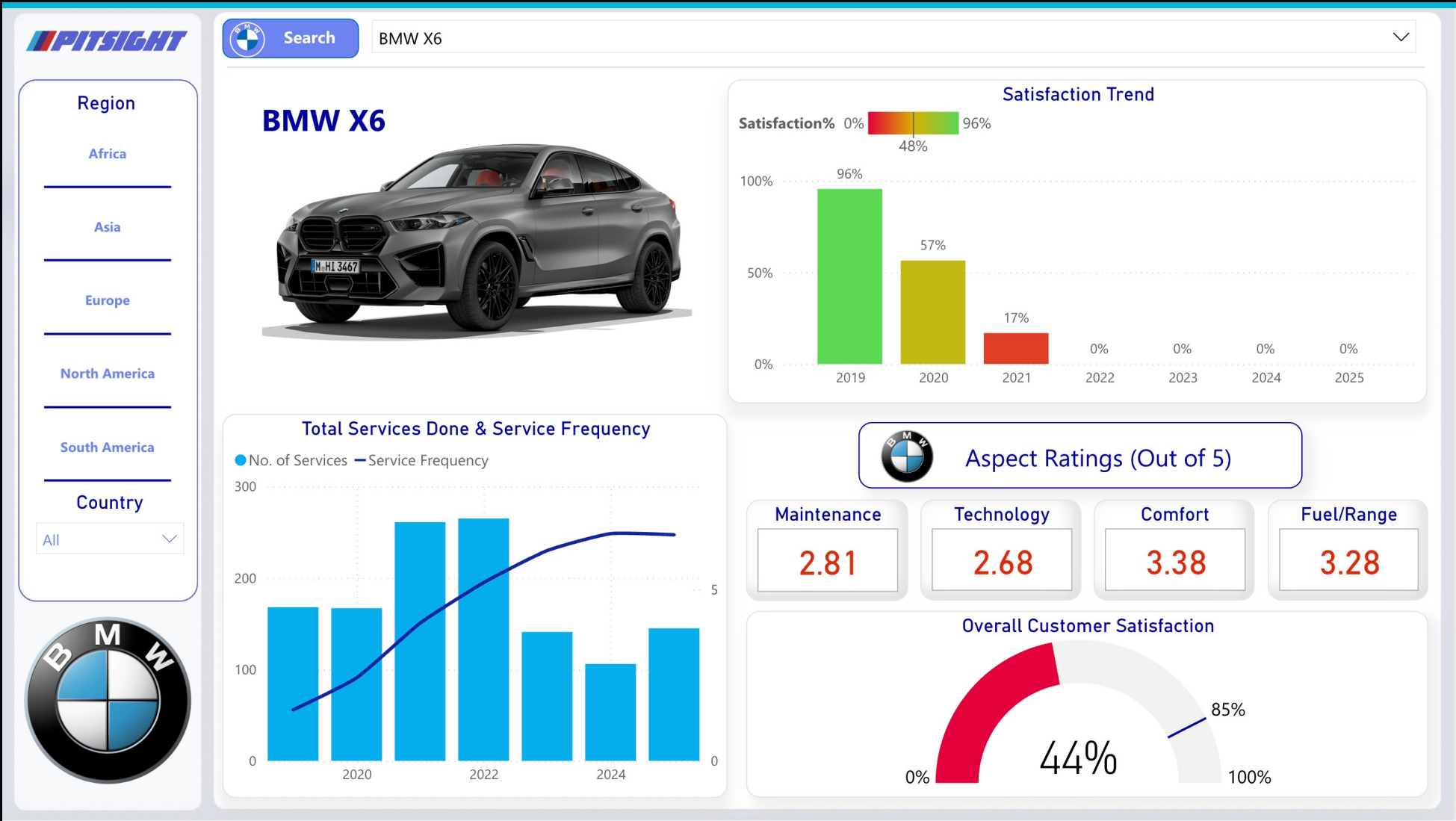
The sales page shows the decline.
The next page shows why.

Average Price: \$82K

Premium segment, buyers expect reliability at this price point.

Sentiment

NLP decodes what 103K customers actually feel about each model



Satisfaction Trend

Conditional bars: green >85%, yellow 50-85%, red <50%.
X6: green in 2019 → red by 2021 → zero.

Service Frequency & Duration

Rising bars + Rising line = reliability collapsing.
Bars show volume, line shows service frequency.

Aspect Health Cards

4 scores out of 5. Maintenance 2.81, Technology 2.68, both red.
Pinpoints exactly what's failing.

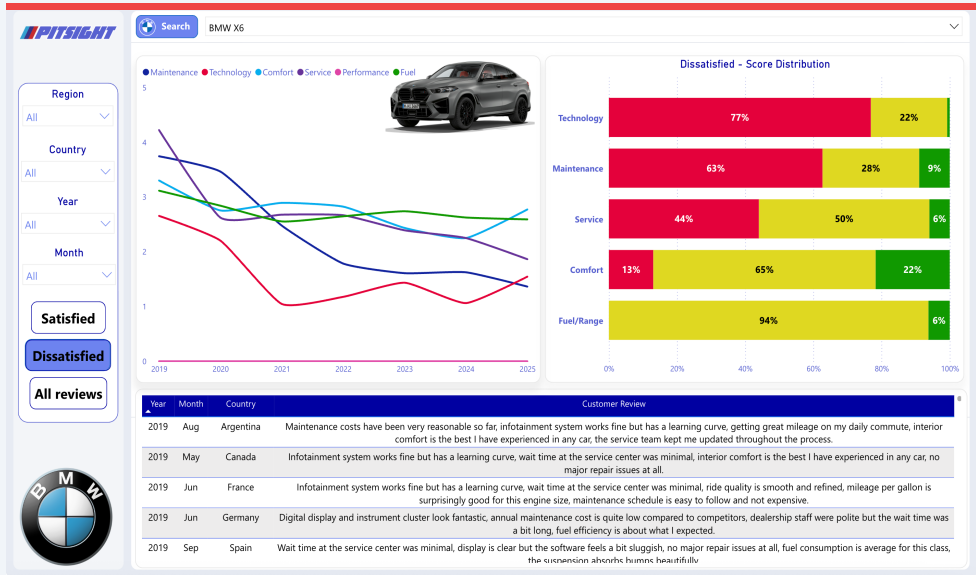
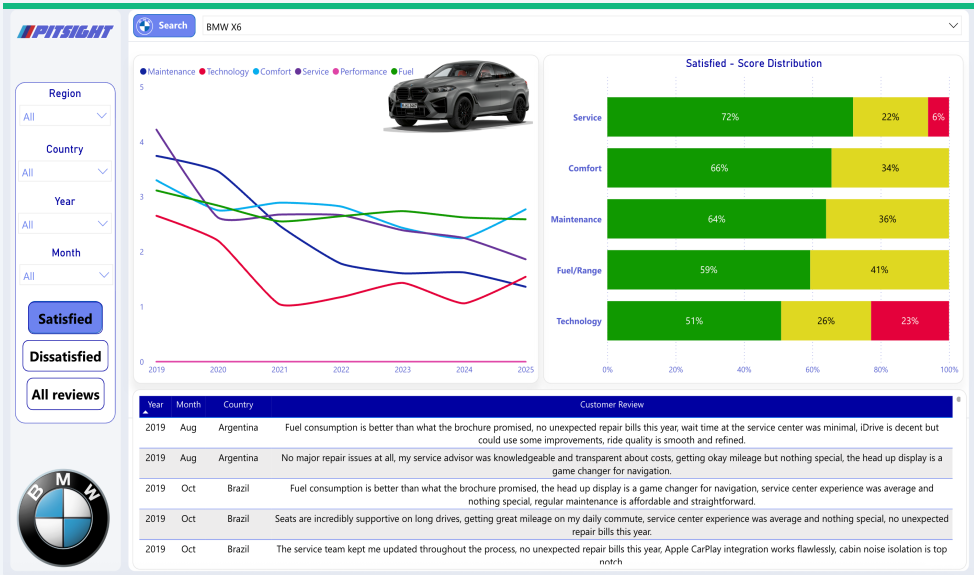
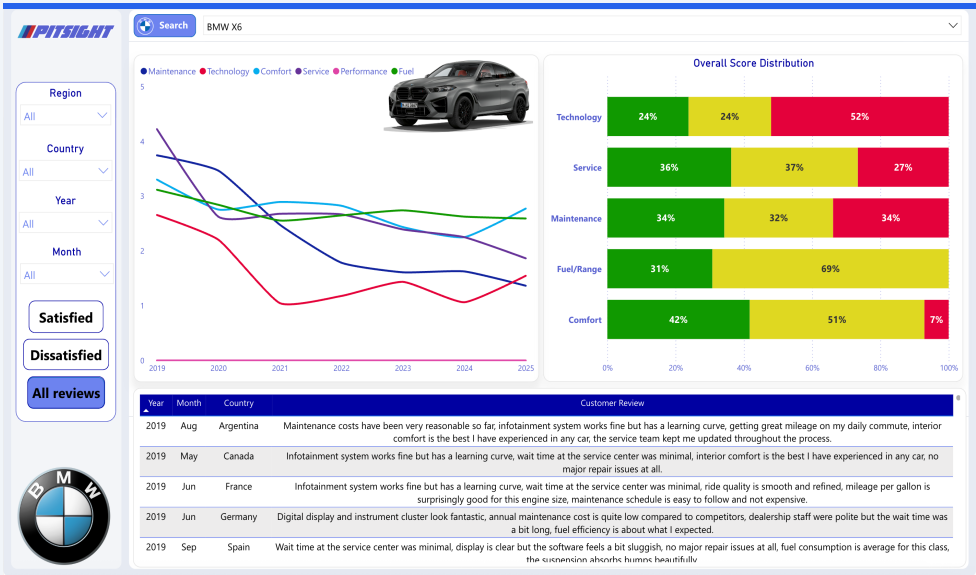
Overall Gauge

44% — deep in the red zone.
One number that summarizes the entire model's health.

Select any model — all visuals update instantly

Reviews

Toggle between Satisfied, Dissatisfied, and All reviews, scores shift dramatically



Overall Score Distribution
across all Customers

What Happy Customers
Value Most

Where the Pain Points
Really are

What This Toggle Reveals

Satisfied X6 owners: Service 72% positive, but Technology only 51%, even happy customers flag tech issues. The aspect line chart shows maintenance (blue line) crashing from 3.8→1.2.

Dissatisfied X6 owners: Technology 77% negative (Surprise! Its Back!) but Fuel/Range 94% neutral-to-negative, Service 50% neutral. Unhappy owners complain about technology, running costs and somewhat about Service.



Service Visit Cliff

78% → 10% satisfaction between 3rd and 4th service visit.
This is the point of no return.

Target: keep every customer below 3 visits per year

Service Duration Impact

95% satisfaction at 3-5 days.
48% at 20+ days in service.
Every extra day costs ~5pp.

Target: complete service within 10 business days

Portfolio Risk

4 models below 80% satisfaction.
X6 (44%), i8 (63%), i3 (69%), X7 (73%).
EVs top the ranking at 96%.

**Intervene on bottom 4.
Replicate EV service model.**

Key Findings

BMW X6

Reliability Crisis

Service visit frequency 1 → 7 services,
Sales – 80%.
Lowest scores in ALL 6 aspects.
Needs immediate intervention or
discontinuation.

**96% → 0%
satisfaction**

BMW X5

Recovery Proof

2022 quality dip fixed in one year.
Service visits 4.0→1.2 after intervention.
Proof that operational fixes work.

**21% → 99%
recovery**

EV Models

Portfolio Leaders

Service visits dropped 65% (2.9→1.0).
i4, iX, iX3 top the ranking.
Fewer Repairs = Happiest Customers.

**54% → 99%
satisfaction**

Technology

Weakest Aspect

Lowest-scoring across ALL segments.
10× more negative than comfort.
iDrive/software lags behind hardware
quality.

**19.4%
negative
reviews**

"How often it broke + how long the service took = 87% of customer satisfaction."

Looking Forward

This framework transfers to any automotive CX program

Product Strategy

Aspect-level scoring shows WHERE to invest. Technology is the weakest link across every segment. A targeted software quality initiative on iDrive would lift the entire portfolio's satisfaction scores.

After-Sales Operations

The service cliff (78%→10% at visit #3) and service duration threshold (15 days) give service teams specific, measurable targets. These aren't opinions, they're data-backed intervention points.

Connected Car Services

The aspect-scoring framework is designed to work on any unstructured text. A natural next step would be applying it to connected car feedback, OTA update reviews, app ratings, service chatbot transcripts.